DBMS LAB – SQL COMMANDS

Each question is followed by separate query and output screenshots.

Name:KUMARES.S.U

Reg no:1912260711

Q1. DDL Commands using CREATE, ALTER, TRUNCATE and DROP

Query Screenshot:

```sql
CREATE DATABASE IF NOT EXISTS dbms_lab;
USE dbms_lab;

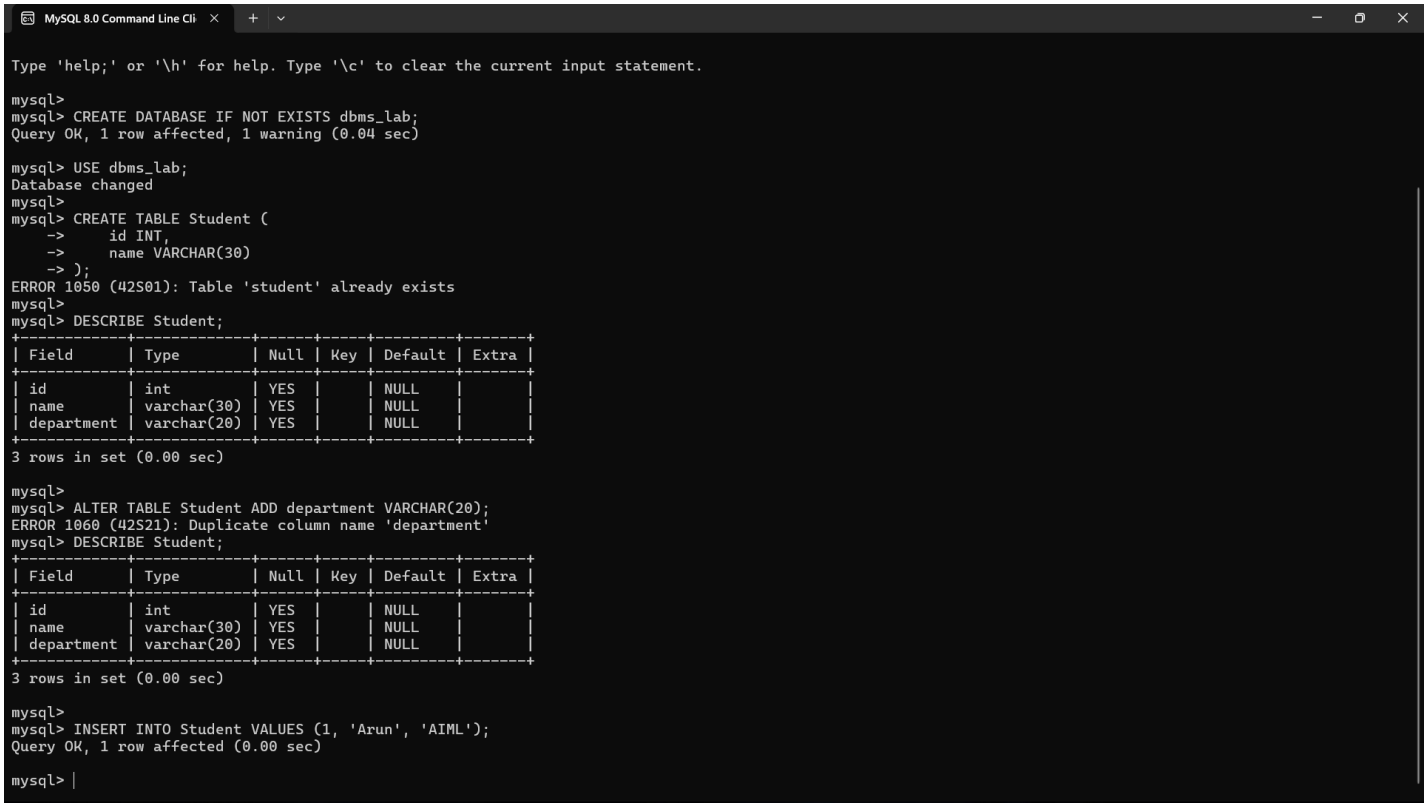
CREATE TABLE Student (
    id INT,
    name VARCHAR(30)
);

DESCRIBE Student;

ALTER TABLE Student ADD department VARCHAR(20);
DESCRIBE Student;

INSERT INTO Student VALUES (1, 'Arun', 'AIML');
```

Output Screenshot



Q2. DDL Commands with Constraints

Query Screenshot:

```sql
CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(30) UNIQUE
);

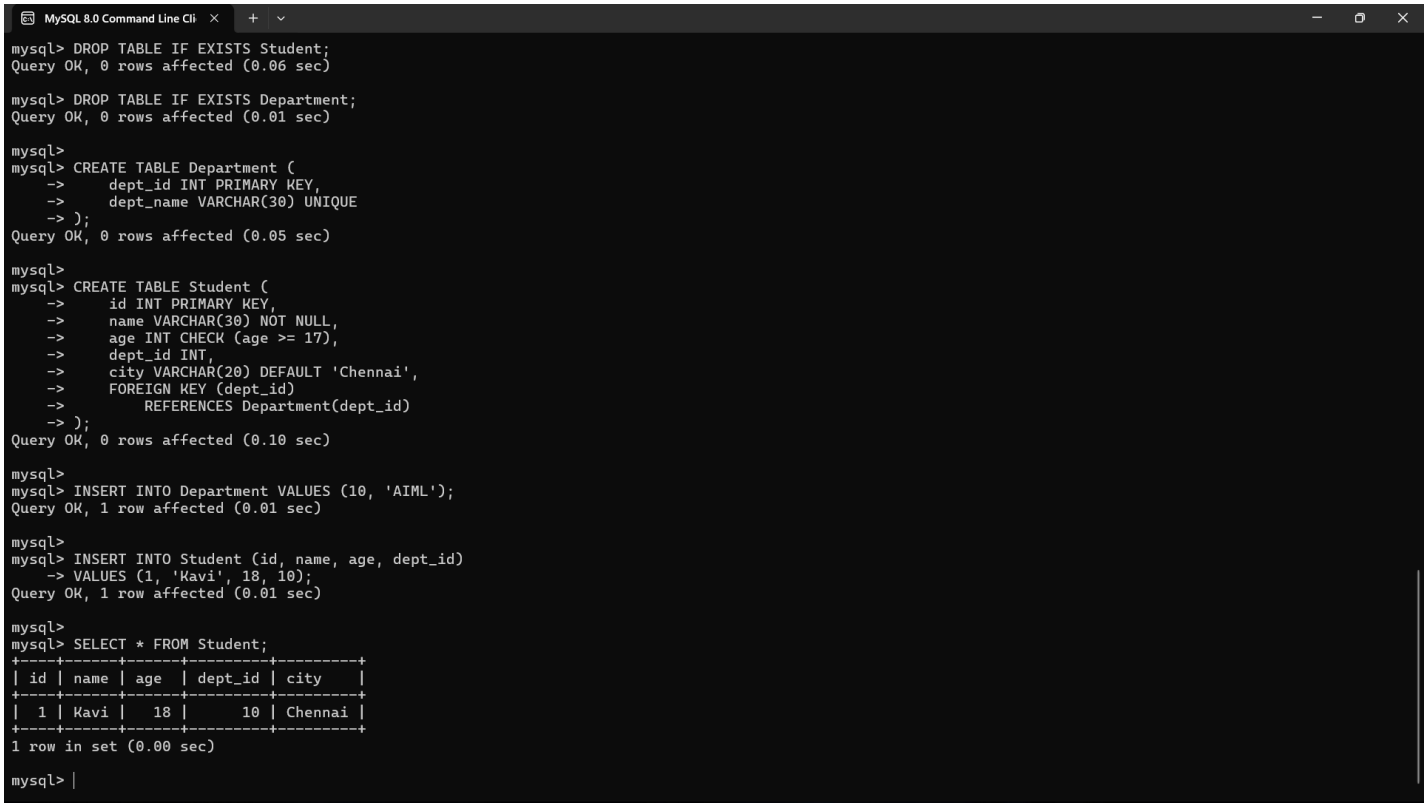
CREATE TABLE Student (
    id INT PRIMARY KEY,
    name VARCHAR(30) NOT NULL,
    age INT CHECK (age >= 17),
    dept_id INT,
    city VARCHAR(20) DEFAULT 'Chennai',
    FOREIGN KEY (dept_id)
        REFERENCES Department(dept_id)
);

INSERT INTO Department VALUES (10, 'AIML');

INSERT INTO Student (id, name, age, dept_id)
VALUES (1, 'Kavi', 18, 10);

SELECT * FROM Student;
```

Output screenshot:



Q3. DML Commands using INSERT and SELECT

Query Screenshot:

```sql
USE dbms_lab;

DROP TABLE IF EXISTS Student;

CREATE TABLE Student (
    id INT,
    name VARCHAR(30),
    course VARCHAR(20)
);

INSERT INTO Student VALUES (1, 'Anu', 'AIML');
INSERT INTO Student VALUES (2, 'Bala', 'CSE');

SELECT * FROM Student;
```

Output Screenshot:

```
mysql>
mysql> DROP TABLE IF EXISTS Student;
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE Student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     course VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> INSERT INTO Student VALUES (1, 'Anu', 'AIML');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Student VALUES (2, 'Bala', 'CSE');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Student;
+------+------+--------+
| id   | name | course |
+------+------+--------+
|    1 | Anu  | AIML   |
|    2 | Bala | CSE    |
+------+------+--------+
2 rows in set (0.00 sec)
```

Q4. DML Commands using UPDATE and DELETE

Query screenshot:

```sql
DROP TABLE IF EXISTS Student;

CREATE TABLE Student (
    id INT,
    name VARCHAR(30),
    marks INT
);

INSERT INTO Student VALUES (1, 'Anu', 75);
INSERT INTO Student VALUES (2, 'Bala', 68);

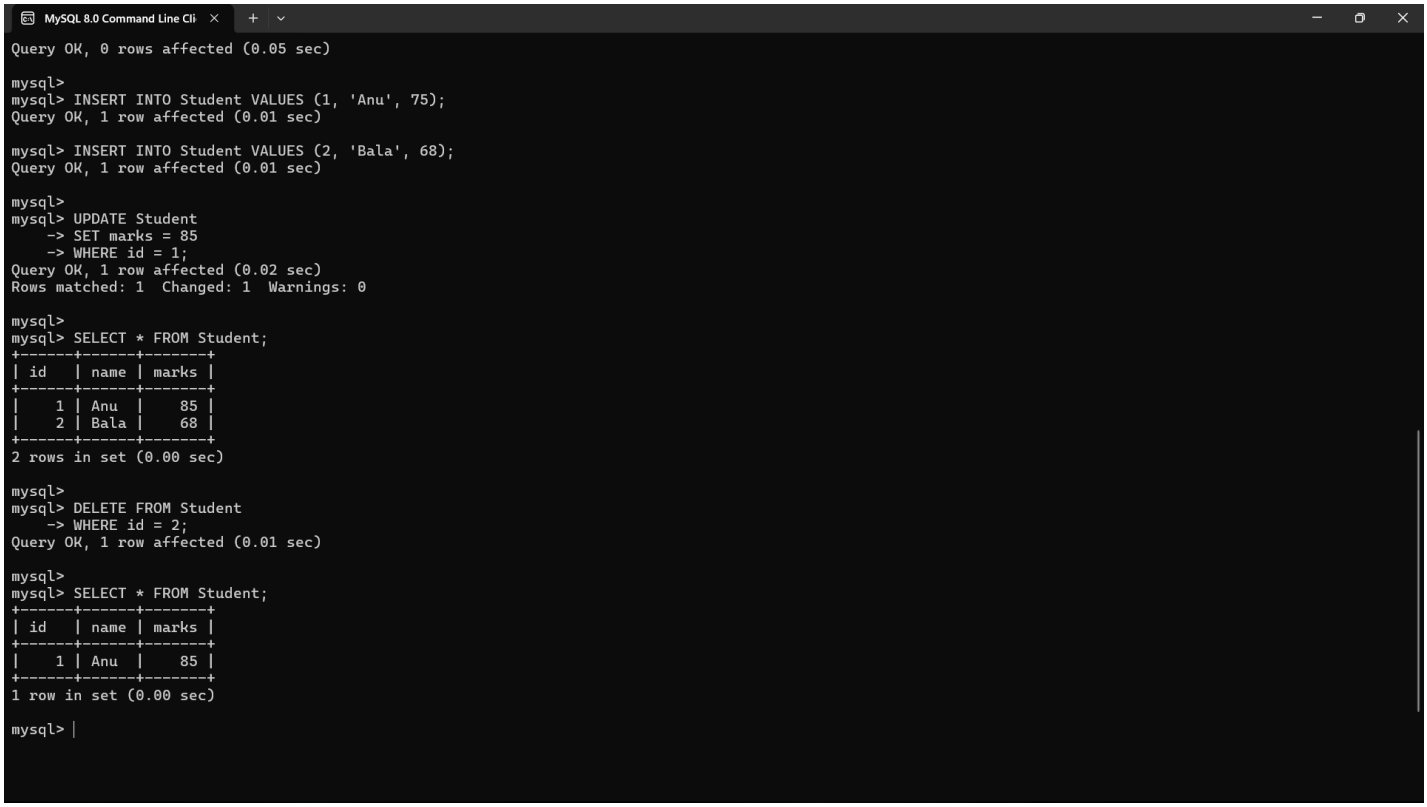
UPDATE Student
SET marks = 85
WHERE id = 1;

SELECT * FROM Student;

DELETE FROM Student
WHERE id = 2;

SELECT * FROM Student;
```

output Screenshot:


MySQL 8.0 Command Line Cli
Query OK, 0 rows affected (0.05 sec)
mysql>
mysql> INSERT INTO Student VALUES (1, 'Anu', 75);
Query OK, 1 row affected (0.01 sec)
mysql> INSERT INTO Student VALUES (2, 'Bala', 68);
Query OK, 1 row affected (0.01 sec)
mysql>
mysql> UPDATE Student
-> SET marks = 85
-> WHERE id = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql>
mysql> SELECT * FROM Student;
+------+------+-------+
| id | name | marks |
+------+------+-------+
| 1 | Anu | 85 |
| 2 | Bala | 68 |
+------+------+-------+
2 rows in set (0.00 sec)
mysql>
mysql> DELETE FROM Student
-> WHERE id = 2;
Query OK, 1 row affected (0.01 sec)
mysql>
mysql> SELECT * FROM Student;
+------+------+-------+
| id | name | marks |
+------+------+-------+
| 1 | Anu | 85 |
+------+------+-------+
1 row in set (0.00 sec)
mysql>

Q5. TCL Commands – COMMIT, SAVEPOINT and ROLLBACK

Query Screenshot:

1

```sql
USE dbms_lab;

DROP TABLE IF EXISTS Account;

CREATE TABLE Account (
    id INT PRIMARY KEY,
    balance INT
) ENGINE = InnoDB;

INSERT INTO Account VALUES (1, 1000);

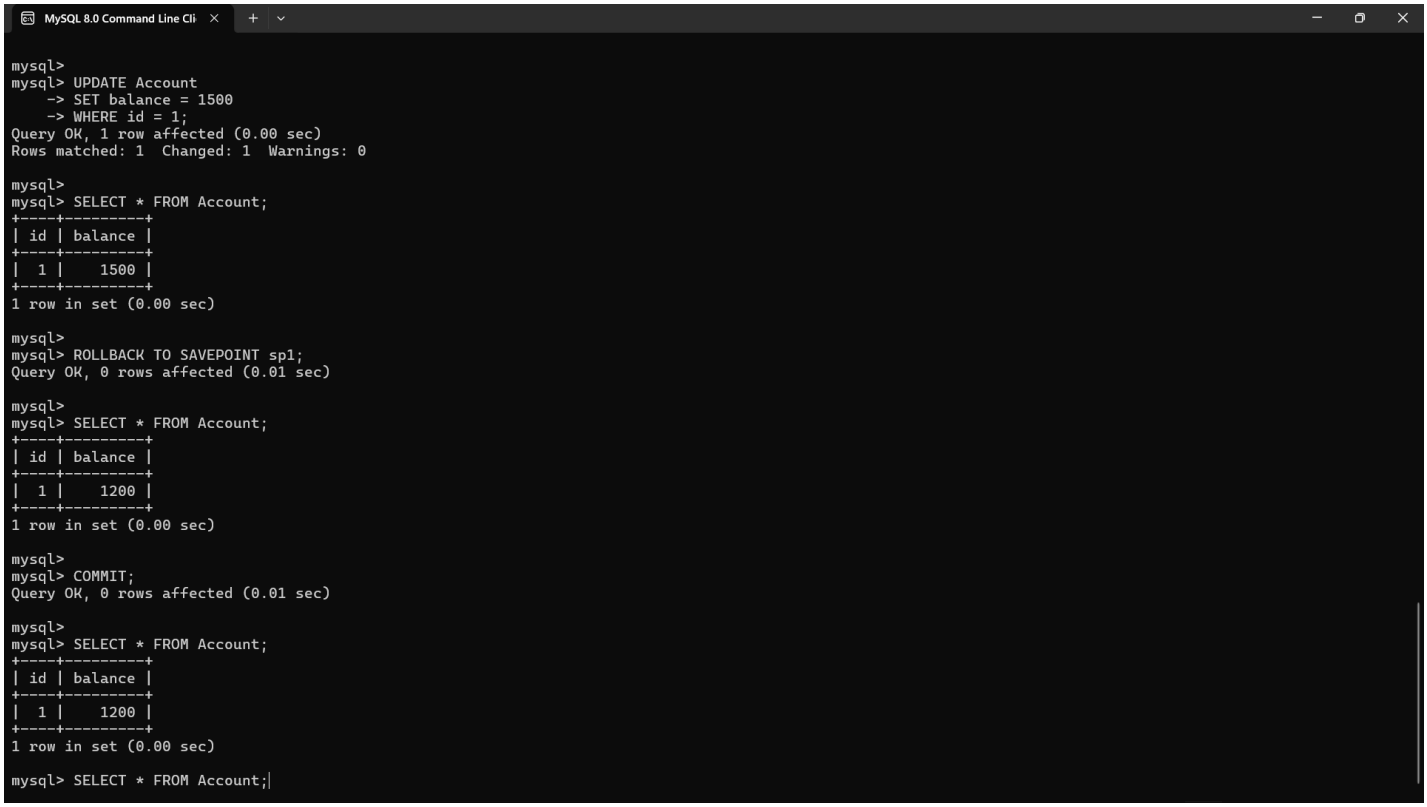
START TRANSACTION;

UPDATE Account
SET balance = 1200
WHERE id = 1;

SAVEPOINT sp1;

UPDATE Account
```

Output Screenshot:


MySQL 8.0 Command Line Cli
mysql>
mysql> UPDATE Account
-> SET balance = 1500
-> WHERE id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
| 1 | 1500 |
+----+---------+
1 row in set (0.00 sec)
mysql>
mysql> ROLLBACK TO SAVEPOINT sp1;
Query OK, 0 rows affected (0.01 sec)
mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
| 1 | 1200 |
+----+---------+
1 row in set (0.00 sec)
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
| 1 | 1200 |
+----+---------+
1 row in set (0.00 sec)
mysql> SELECT * FROM Account;